Tertiary syphilis

- After several years, manifestations of tertiary syphilis appear. These consist of cardiovascular lesions including aneurysms (enlargement of an artery), gummata (a small rubbery granuloma that has a necrotic centre) and meningovascular manifestations. Tertiary lesions contain few spirochetes.
- In few cases, neurosyphilis such as tabesdorsalis or general paralysis of the insane develops. These are known as late tertiary or quaternary syphilis.

Congenital syphilis

In congenital syphilis, the infection is transmitted from mother to fetus by crossing the placental barrier.

Non – Venereal syphilis

It may occur in doctors or nurses due to contact with patients lesion during examination. The primary chancre occurs usually on the fingers.

7.12.4 Laboratory Diagnosis

The diagnosis of syphilis includes

a. Demonstration of Treponemes

b. Serological tests

Specimen: Exudates are collected from the chancre. Blood (serum) is collected for serology.

Demonstration of Treponemes

a. **Dark ground microscopy:** The wet film is prepared with exudates and examined under dark ground microscope. Under dark field examination *Treponema pallidum* appears motile spiral organism.

b. **Treponemes in tissues:** It can be demonstrated by silver impregnation method of staining.

Serological tests

Non – Treponemal tests – In the standard tests for syphilis includes;

a. VDRL – Venereal Diseases Research Laboratory test.

b. RPR – Rapid Plasma Reagain (Figure 7.23).

VDRL or RPR tests are used for serological screening for syphilis and more useful for the assessment of cure following treatment.



TRUST – Toluidine red unheated serum test, modified form of RPR test.

Treponemal Tests: The treponemal tests includes

a. TPHA – Treponema pallidum hemagglutination assay

b. FTA –ABS – Fluorescent treponemal antibody absorption test.

These two tests are used to confirm the diagnosis.



Figure 7.23: Rapid Plasma Reagain test

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